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RESEARCH INTERESTS

Reliable and verifiable large language model (LLM) systems for clinical knowledge; evaluation of medical AI beyond outcome accuracy (evidence grounding, reasoning faithfulness, appropriate abstention); retrieval-augmented generation for non-English clinical and educational content; implementation of AI in resource-constrained health systems.

EDUCATION

Doctor of Medicine (Doctor en Medicina) Sep 2020 – Jul 2026
Universidad de Ciencias Médicas de La Habana, Facultad de Ciencias Médicas "Julio Trigo López", Havana, Cuba

- **GPA 4.80 / 5.00** (801 credits); State Examination 5 / 5. Degree conferred 18 July 2026.
- **Graduated with Título de Oro (Summa Cum Laude)**, the university's highest academic distinction.
- Six-year programme taught in Spanish; final-year clinical rotations in Internal Medicine, Paediatrics, General Surgery, Obstetrics & Gynaecology and Comprehensive General Medicine (all graded 5 / 5).
- Full Government of Ghana scholarship (Ghana Scholarships Secretariat), 2020–2026.

BSc Computer Engineering programme (not completed) Sep 2018 – 2020
Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

- Completed three semesters (66 credits; cumulative weighted average 72.58 / 100). Programme discontinued in 2020 on beginning medical studies in Cuba; final-semester courses were not completed.
- Relevant coursework: Algebra (A); Calculus with Analysis (A); Differential Equations (B); Discrete Structures (A); Programming and Problem Solving (A); Data Structures and Algorithms (A); Computer Organization (A); Introduction to MATLAB and LabVIEW (A).

RESEARCH EXPERIENCE

Student Investigator and Project Originator Jan 2025 – present
Faculty of Medical Sciences "Julio Trigo López" and Hospital Clínico Quirúrgico Docente "Julio Trigo López", Havana
Supervisors: Prof. Dr. C. Juan Fundora Lliteras (research methodologist, Profesor Titular); Dr. Armando Jonathan Pérez López (Internal Medicine)

- **Barriers to AI-assisted diagnosis in Cuban health institutions (mixed-methods study).** Formulated the research question and designed a cross-sectional mixed-methods study combining a national online survey of clinical staff with semi-structured interviews at Hospital Julio Trigo López.
- Developed the survey instrument, interview guide, informed-consent form and study summary; prepared the ethics submission. Ethics approval obtained; data collection is in progress in Cuba, conducted by the research team.
- Presented the study protocol and conceptual framework at two faculty and hospital scientific meetings in 2025 (Premio Mención; Relevant Project Award).
- **Institutional R&D project.** The study was expanded into the faculty's three-year R&D project *La formación médica y la asistencia con aplicación de la Inteligencia Artificial* (Medical Training and Clinical Care with Artificial Intelligence; Jan 2026 – Dec 2028; PI: Prof. Fundora), with Hospital Julio Trigo López and the Arroyo Naranjo Municipal Health Directorate as participating institutions. Originated the project proposal, as confirmed by the PI.
- Responsible within the project for designing and deploying its AI-enabled web platform for students (MedEstudia, below); co-author of the project's first manuscript (under review).

Independent Evaluation Audit of a Multi-Agent Diagnostic LLM Benchmark Oct 2026
Independent, unsupervised work on the public GP-EVAL benchmark (Cao Lab, Monash University). Code and findings: github.com/laseeru/gpeval-audit

- Developed a Python audit of the benchmark's public results database (4,056 model runs across 312 simulated primary-care cases). Identified 3 runs scored as correct although the grader returned "NO" (a substring-parsing defect) and 46 runs that ended without any grader verdict.
- Applied a heuristic screen for answer leakage that flagged about 100 of the 312 cases in which requestable test results may disclose the target diagnosis. The screen is deliberately over-inclusive and requires case-by-case clinical review.

- Built a reproducible test harness (pinned environment, database integrity checks, API-call guard, per-run JSON logs) and re-ran three cases with current models. Physician review of the transcripts found reasoning errors not detected by outcome-only grading, such as a model citing investigations it had never requested.
- Discussed the findings with the developing lab, which has since archived the benchmark; the audit code and a physician-reviewed analysis were released publicly with the lab's agreement (Nov 2026).

RESEARCH SOFTWARE AND TECHNICAL PROJECTS

MedEstudia: AI-assisted study platform for Spanish-speaking medical students Nov 2025 – present
Sole developer. Platform component of the institutional R&D project above.

- Bilingual (Spanish/English) web application (React, TypeScript, serverless API) that uses DeepSeek-V3.1 via Azure AI Foundry to generate multiple-choice questions, quizzes, topic explanations and structured clinical-guideline summaries; includes an error notebook, timed study sessions and 141 automated tests.
- Built a retrieval-augmented generation (RAG) pipeline over a curated Spanish-language corpus of 38 Cuban medical textbooks (42,572 text segments after filtering corrupted OCR output): Azure OpenAI embeddings, Supabase pgvector (HNSW) search, inline passage citations, server-side removal of citations that point to no retrieved passage, and an explicit "no verifiable sources" flag when nothing relevant is found.
- Designed the accompanying evaluation (no-retrieval baseline vs. abstention instruction vs. retrieval with citations; retrieval Recall@k and physician-graded accuracy and faithfulness). A pilot already exposed a cited but unfaithful claim, motivating faithfulness rather than citation presence as the outcome. The evaluation itself has not yet been run.
- Preliminary usability pilot with 15 volunteer medical students (qualitative feedback); presented at the XXXIV Jornada Científica Estudiantil, 2026.
- The competition prototype included an embedding-based retrieval module over Cuban Ministry of Public Health clinical protocols. **3rd place, Monash Global Medical AI Challenge 2025** (solo entrant).

Faculty Research Assistant: retrieval-augmented question answering Aug 2026
Sole developer; for the Faculty of Medical Sciences "Julio Trigo López". Phase 1 (command-line) prototype.

- Python pipeline that answers students' questions about thesis, methodology and ethics regulations from the faculty's own documents: PDF ingestion and cleaning, chunking, local multilingual embeddings (multilingual-e5-small), ChromaDB vector store, and DeepSeek generation.
- Returns page-level citations and refuses to answer when the retrieved evidence is insufficient; covered by pytest unit tests. A formal retrieval and answer-accuracy evaluation has not yet been conducted.

Online platform for the Convención Científica "Julio Trigo López" 2026 2026
Built at the faculty's request; awarded a Certificate of Recognition by the Dean.

- Online system for the faculty's 2026 scientific convention, handling abstract registration and qualification, review commissions, and certificate lookup and generation for approximately 180 participants and 390 participant–abstract records.

GMDC Prep: licensing-examination practice application Aug – Sep 2025
Sole developer. Prototype.

- Web application for candidates preparing for the Ghana Medical and Dental Council licensing examination: topic-based practice, progress tracking and email (magic-link) authentication (Next.js, Prisma, PostgreSQL). No AI component.

Selected software for external clients 2026

- **APC-ELI website**: bilingual website for the English Language Teaching Section of the Association of Pedagogues of Cuba (Next.js, internationalisation).
- **HGS Business Manager**: public website and owner-only inventory administration system for a horticultural trading enterprise in Tamale, Ghana (Next.js, PostgreSQL, Prisma, authentication).

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed article

1. **Abdulai ARD.** Realidad y ensueño: travesía a través de la confusión onírica [Reality and dream: a journey through dream–reality confusion]. *Ciencia&Conciencia* (student scientific journal, Escuela Latinoamericana de Medicina). 2023;6(1):83–87. Review article. revcienciaconciencia.sld.cu/…/article/view/77

Manuscripts under review

2. Fundora Lliteras J, Betancourt Torres I, Mena Mugica R, **Abdulai ARD**. La inteligencia artificial en la educación desde la concepción histórico-cultural de Vygotsky hacia una mediación contextualizada [Artificial intelligence in education from Vygotsky's historical-cultural perspective: towards contextualised mediation]. *Revista Varona*. Under review.

3. **Abdulai ARD.** Inteligencia artificial en medicina materno-fetal: avances y desafíos en el diagnóstico y tratamiento [Artificial intelligence in maternal–fetal medicine: advances and challenges in diagnosis and treatment]. *Ciencia&Conciencia*. Narrative review. Under review.

Conference papers (XXXIV Jornada Científica Estudiantil, virtual, FCM "Julio Trigo López", 2026)

4. **Abdulai ARD**, Ahmed A. MedEstudia: plataforma educativa para razonamiento clínico asistido por inteligencia artificial [MedEstudia: an educational platform for AI-assisted clinical reasoning].
5. **Abdulai ARD**, Ahmed A. Inteligencia artificial generativa para crear herramientas educativas en contextos de bajos recursos [Generative AI for building educational tools in low-resource settings].
6. **Abdulai ARD.** Percepción estudiantil sobre inteligencia artificial guiada en educación médica [Student perceptions of guided artificial intelligence in medical education].

CONFERENCE PRESENTATIONS

1. **Abdulai ARD.** Vygotsky y la inteligencia artificial: el docente como creador de herramientas mediadoras. Convención Científica "Julio Trigo López" 2026, session on ICT-based learning management, Havana, 2026.
2. **Abdulai ARD.** Three papers on MedEstudia and AI in medical education (listed above). XXXIV Jornada Científica Estudiantil (virtual), Havana, 2026.
3. **Abdulai ARD.** MedEstudia: an AI study assistant for Cuban medical students. Final pitch, Monash Global Medical AI Challenge (online), Dec 2025. 3rd place.
4. **Abdulai ARD.** Barriers to implementing AI in low-resource healthcare. Delegate speech, Youth Vision Assembly 2025, Amsterdam, Netherlands, 19–22 Sep 2025.
5. **Abdulai ARD.** Barreras y oportunidades para la implementación de diagnósticos médicos impulsados por IA en instituciones cubanas [Barriers and opportunities for implementing AI-driven medical diagnostics in Cuban institutions]. Jornada Científica Hospital "Julio Trigo López" / Jornada de Residentes, Havana, Jun 2025. Relevant Project Award.
6. **Abdulai ARD.** La inteligencia artificial, la formación médica y la asistencia: barreras y oportunidades [AI, medical training and clinical care: barriers and opportunities]. Jornada Científica Estudiantil, FCM "Julio Trigo López", Havana, 3 Jun 2025. Premio Mención.

AWARDS AND DISTINCTIONS

2026	**Título de Oro (Summa Cum Laude)**, Doctor of Medicine, Universidad de Ciencias Médicas de La Habana
2026	**Certificate of Recognition** for outstanding contribution to the development of the Convención Científica 2026, conferred by the Dean, Faculty of Medical Sciences "Julio Trigo López"
2025	**3rd Place**, Monash Global Medical AI Challenge (Cao Lab and Nexus, Monash University; 159 participants from 19 countries)
2025	**Relevant Project Award**, Jornada Científica Hospital "Julio Trigo López" / Jornada de Residentes
2025	**Premio Mención**, Jornada Científica Estudiantil, FCM "Julio Trigo López"
2025	**Fully Funded Delegate**, Youth Vision Assembly, Amsterdam (Centre for Diplomatic Advancement; competitive selection)
2025	Selected participant with fee waiver, Neuromatch Academy Computational Neuroscience; unable to attend because of platform access restrictions in Cuba
2020–2026	**Government of Ghana Scholarship** to study medicine in Cuba, Ghana Scholarships Secretariat (tuition, maintenance and travel)

CLINICAL EXPERIENCE

Pre-registration Clinical Attachment Nov 2026 – present
37 Military Hospital, Accra, Ghana

- Supervised clinical attachment required for registration with the Ghana Medical and Dental Council; licensing examination scheduled for June 2027.

Clinical clerkships (undergraduate) 2022 – 2026
Teaching hospitals and polyclinics affiliated with the Universidad de Ciencias Médicas de La Habana

- Rotations in Internal Medicine, Surgery, Paediatrics, Obstetrics & Gynaecology, Emergency Medicine, Psychiatry, Comprehensive General (Family) Medicine and surgical subspecialties.

TECHNICAL SKILLS

Programming	Python (data pipelines, retrieval systems, evaluation scripts); TypeScript/JavaScript (React, Next.js); SQL (PostgreSQL, SQLite); MATLAB (undergraduate coursework).

AI and ML	LLM application development (DeepSeek, Azure AI Foundry); retrieval-augmented generation (Azure OpenAI and sentence-transformer embeddings; pgvector and ChromaDB vector search; citation grounding and refusal logic); LLM evaluation harnesses; supervised machine learning fundamentals (regression, classification; scikit-learn, coursework level).
Software practice	Git/GitHub; automated testing (pytest, Vitest); Prisma, Supabase; serverless deployment (Vercel); AI-assisted development tools.
Research methods	Mixed-methods study design; survey and interview-guide development; informed consent and ethics submissions; literature review; biostatistics (coursework).

RELEVANT COURSEWORK

Medicine (Havana)	Biostatistics (5); Research Methodology (5); Medical Genetics (5); Molecular Biology (5); Public Health (5); Health Promotion (5); Medical Microbiology and Parasitology (5); Science, Technology and Society (exempted).
Engineering (KNUST)	Algebra; Calculus with Analysis; Differential Equations; Discrete Structures; Programming and Problem Solving; Data Structures and Algorithms; Computer Organization; Basic Electronics; Circuit Theory.

Grading scales: Havana 5 = Excellent, 4 = Good, 3 = Fair; KNUST letter grades A–D.

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

Oct 2026	**Microsoft Certified: Azure AI Fundamentals (exam AI-901)**, Microsoft; score 933 / 1000
Aug 2025	**Python for Everybody Specialization (5 courses)**, University of Michigan, Coursera
Apr 2025	**Supervised Machine Learning: Regression and Classification**, DeepLearning.AI and Stanford University, Coursera
Mar 2025	**Artificial Intelligence for Breast Cancer Detection**, Johns Hopkins University, Coursera

Coursera items are online, non-credit courses.

LANGUAGES

English (fluent; language of secondary and university engineering education); **Spanish** (full professional proficiency; six-year medical degree taught in Spanish); **Dagbani** (native); **Gonja** (native).